Movie_Genre_Classification - C | CodSoft_Task3_CustomerChurn | Spam_SMS_Detection.ipynb - C | Ask Gemini

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spam.csv

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[2] 5s

```
!pip install wordcloud
```

```
Requirement already satisfied: wordcloud in /usr/local/lib/python3.12/dist-packages (1.9.6)
Requirement already satisfied: numpy>=1.19.3 in /usr/local/lib/python3.12/dist-packages (from wordcloud) (2.0.2)
Requirement already satisfied: pillow in /usr/local/lib/python3.12/dist-packages (from wordcloud) (11.3.0)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (from wordcloud) (3.10.0)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->wordcloud) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib->wordcloud) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib->wordcloud) (4.62.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->wordcloud) (1.5.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib->wordcloud) (26.1)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->wordcloud) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib->wordcloud) (2.9.0.p
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib->wordcloud
```

[4] 2m

```
import pandas as pd
import numpy as np
import re
import os
import joblib
import matplotlib.pyplot as plt

from wordcloud import WordCloud

from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
```

Variables Terminal

Python 3

ENG IN 11:03 12-05-2026

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shreenithy_task2_predictions.csv

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[4] 2m

```
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    classification_report,
    confusion_matrix,
    ConfusionMatrixDisplay,
    roc_curve,
    auc
)


print("Uploaded files:")
print(os.listdir())
file_name = "spam.csv"
try:
    sms_data = pd.read_csv(file_name, encoding="latin-1")
except:
    sms_data = pd.read_csv(file_name, sep="\t", names=["Label", "Message"], encoding="latin-1")

print("\nDataset loaded successfully!")
print(sms_data.head())
print("\nOriginal Columns:")
print(sms_data.columns)

if "v1" in sms_data.columns and "v2" in sms_data.columns:
    sms_data = sms_data[["v1", "v2"]]
    sms_data.columns = ["Label", "Message"]
```

Variables Terminal

Python 3

ENG IN 11:04 12-05-2026

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shreenithy_spam_tfidf_vectorizer...

shreenithy_task2_model_compar...

shreenithy_task2_predictions.csv

spam.csv

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[4] 2m

```
    elif len(sms_data.columns) >= 2:
        sms_data = sms_data.iloc[:, :2]
        sms_data.columns = ["Label", "Message"]

    print("\nFinal Dataset:")
    print(sms_data.head())

    print("\nDataset Shape:", sms_data.shape)

    print("\nMissing Values:")
    print(sms_data.isnull().sum())

    sms_data.dropna(inplace=True)

    print("\nSpam and Ham Count:")
    print(sms_data["Label"].value_counts())

    sms_data["Message_Length"] = sms_data["Message"].apply(len)

    print("\nAverage Message Length:")
    print(sms_data.groupby("Label")["Message_Length"].mean())

    plt.figure(figsize=(6, 4))
    sms_data["Label"].value_counts().plot(kind="bar")
    plt.title("Spam vs Ham SMS Count")
    plt.xlabel("Message Type")
    plt.ylabel("Count")
    plt.show()

    def clean_message(text):
```

Variables Terminal Python 3

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Action required

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shreenithy_task2_model_compar...

shreenithy_task2_predictions.csv

spam.csv

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[4] 2m

```
def clean_message(text):
    text = str(text).lower()
    text = re.sub(r"http\S+", "", text)
    text = re.sub(r"www\S+", "", text)
    text = re.sub(r"[^a-zA-Z\s]", "", text)
    text = re.sub(r"\s+", " ", text).strip()
    return text

sms_data["Cleaned_Message"] = sms_data["Message"].apply(clean_message)

print("\nOriginal Message Example:")
print(sms_data["Message"].iloc[0])

print("\nCleaned Message Example:")
print(sms_data["Cleaned_Message"].iloc[0])

sms_data["Label"] = sms_data["Label"].astype(str).str.lower()

sms_data["Encoded_Label"] = sms_data["Label"].map({
    "ham": 0,
    "spam": 1
})

sms_data.dropna(subset=["Encoded_Label"], inplace=True)

sms_data["Encoded_Label"] = sms_data["Encoded_Label"].astype(int)
X = sms_data["Cleaned_Message"]
y = sms_data["Encoded_Label"]
```

Variables Terminal

Python 3

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[4] ✓ 2m

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

print("\nTraining Size:", X_train.shape[0])
print("Testing Size:", X_test.shape[0])

tfidf_vectorizer = TfidfVectorizer(
    stop_words="english",
    max_features=8000,
    ngram_range=(1, 2),
    min_df=1
)

X_train_vector = tfidf_vectorizer.fit_transform(X_train)
X_test_vector = tfidf_vectorizer.transform(X_test)

print("\nTF-IDF Vectorization completed!")
print("Vector Shape:", X_train_vector.shape)

models = {
    "Naive Bayes": MultinomialNB(),
    "Logistic Regression": LogisticRegression(max_iter=2000),
    "Linear SVM": LinearSVC(),
    "Random Forest": RandomForestClassifier(n_estimators=200, random_state=42)
```

Variables Terminal

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[4] 2m

```
results = {}
predictions = {}

for model_name, model in models.items():
    print("\n====================================")
    print("Training Model:", model_name)
    print("====================================")

    model.fit(X_train_vector, y_train)

    y_pred = model.predict(X_test_vector)

    accuracy = accuracy_score(y_test, y_pred)
    precision = precision_score(y_test, y_pred)
    recall = recall_score(y_test, y_pred)
    f1 = f1_score(y_test, y_pred)

    results[model_name] = {
        "Accuracy": accuracy,
        "Precision": precision,
        "Recall": recall,
        "F1 Score": f1
    }

    predictions[model_name] = y_pred

    print("Accuracy:", accuracy)
    print("Precision:", precision)
    print("Recall:", recall)
```

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[4] 2m

```
comparison_df = pd.DataFrame(results).T

print("\n========== MODEL COMPARISON ==========")
print(comparison_df)

comparison_df.to_csv("shreenithy_task2_model_comparison.csv")

best_model_name = comparison_df["Accuracy"].idxmax()
best_model = models[best_model_name]
best_prediction = predictions[best_model_name]

print("\nBest Model:", best_model_name)
print("Best Accuracy:", comparison_df.loc[best_model_name, "Accuracy"])
plt.figure(figsize=(8, 5))
plt.bar(comparison_df.index, comparison_df["Accuracy"])
plt.title("Spam SMS Detection - Accuracy Comparison")
plt.xlabel("Models")
plt.ylabel("Accuracy")
plt.xticks(rotation=20)
plt.show()
comparison_df[["Precision", "Recall", "F1 Score"]].plot(kind="bar", figsize=(9, 5))
plt.title("Precision, Recall and F1 Score Comparison")
plt.xlabel("Models")
plt.ylabel("Score")
plt.xticks(rotation=20)
plt.show()

cm = confusion_matrix(y_test, best_prediction)

disp = ConfusionMatrixDisplay(
```

Variables Terminal

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[4] 2m

```
)

disp.plot()
plt.title("Confusion Matrix - Best Spam Detection Model")
plt.show()

if best_model_name == "Linear SVM":
    score_values = best_model.decision_function(X_test_vector)
else:
    score_values = best_model.predict_proba(X_test_vector)[:, 1]

fpr, tpr, threshold = roc_curve(y_test, score_values)
roc_auc = auc(fpr, tpr)

plt.figure(figsize=(6, 5))
plt.plot(fpr, tpr, label="AUC = %.2f" % roc_auc)
plt.plot([0, 1], [0, 1], linestyle="--")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve - Spam SMS Detection")
plt.legend()
plt.show()
spam_words = " ".join(sms_data[sms_data["Encoded_Label"] == 1]["Cleaned_Message"])
ham_words = " ".join(sms_data[sms_data["Encoded_Label"] == 0]["Cleaned_Message"])

spam_wordcloud = WordCloud(
    width=800,
    height=400,
    background_color="white"
) generate(spam_words)
```

Variables Terminal

Python 3

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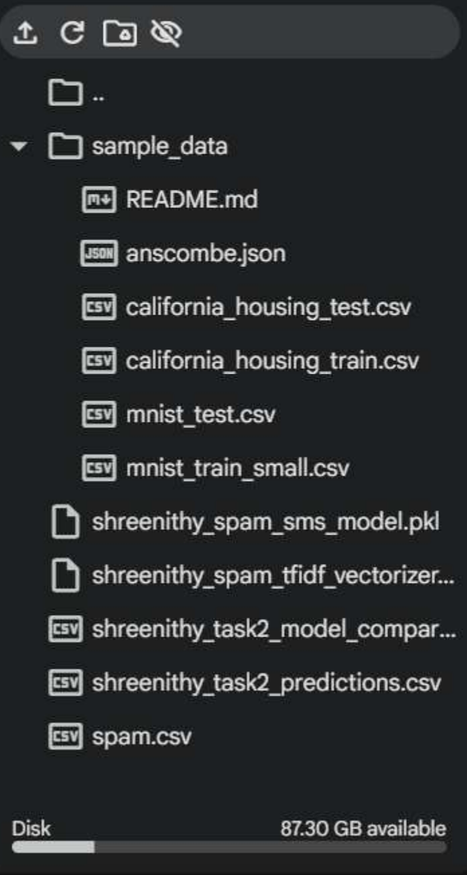

[4] ✓ 2m

```
plt.figure(figsize=(10, 5))
plt.imshow(spam_wordcloud)
plt.axis("off")
plt.title("Most Common Words in Spam Messages")
plt.show()

ham_wordcloud = WordCloud(
    width=800,
    height=400,
    background_color="white"
).generate(ham_words)

plt.figure(figsize=(10, 5))
plt.imshow(ham_wordcloud)
plt.axis("off")
plt.title("Most Common Words in Ham Messages")
plt.show()

prediction_df = pd.DataFrame({
    "Actual_Label": y_test.values,
    "Predicted_Label": best_prediction
})

prediction_df["Actual_Label"] = prediction_df["Actual_Label"].map({
    0: "Ham",
    1: "Spam"
})

prediction_df["Predicted_Label"] = prediction_df["Predicted_Label"].map({
    0: "Ham",
```

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```
Uploaded files:
['.config', 'shreenithy_task2_model_comparison.csv', 'shreenithy_task2_predictions.csv', 'shreenithy_spam_sms_model.pkl', 'shreenit

Dataset loaded successfully!
     v1                                                 v2 Unnamed: 2  \
0   ham  Go until jurong point, crazy.. Available only ...        NaN
1   ham                      Ok lar... Joking wif u oni...        NaN
2  spam  Free entry in 2 a wkly comp to win FA Cup fina...        NaN
3   ham  U dun say so early hor... U c already then say...        NaN
4   ham  Nah I don't think he goes to usf, he lives aro...        NaN

  Unnamed: 3 Unnamed: 4
0        NaN        NaN
1        NaN        NaN
2        NaN        NaN
3        NaN        NaN
4        NaN        NaN

Original Columns:
Index(['v1', 'v2', 'Unnamed: 2', 'Unnamed: 3', 'Unnamed: 4'], dtype='object')

Final Dataset:
  Label                                            Message
0   ham  Go until jurong point, crazy.. Available only ...
1   ham                      Ok lar... Joking wif u oni...
2  spam  Free entry in 2 a wkly comp to win FA Cup fina...
3   ham  U dun say so early hor... U c already then say...
4   ham  Nah I don't think he goes to usf, he lives aro...

Dataset Shape: (5572, 2)

Missing Values:
```

Variables Terminal Python 3

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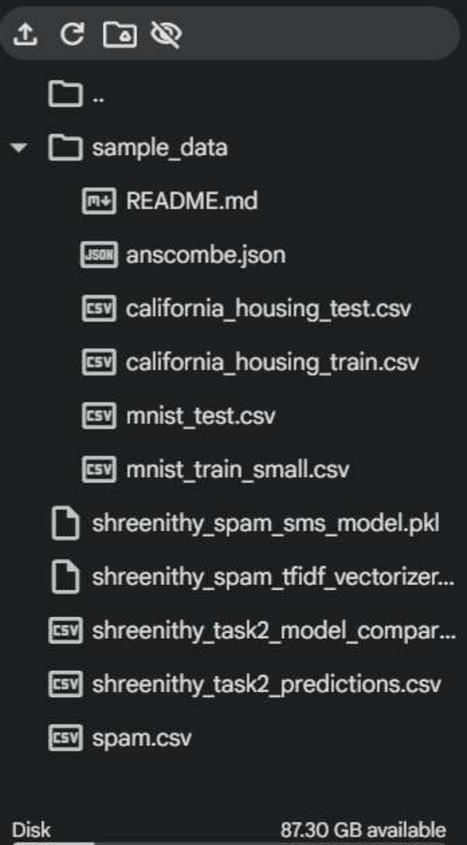
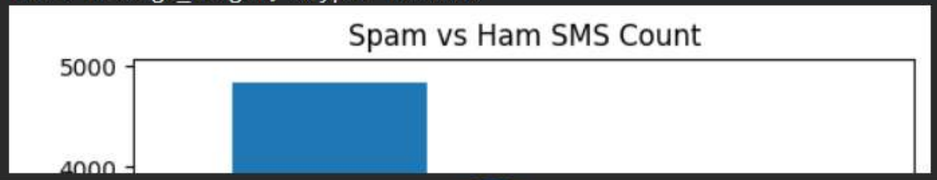
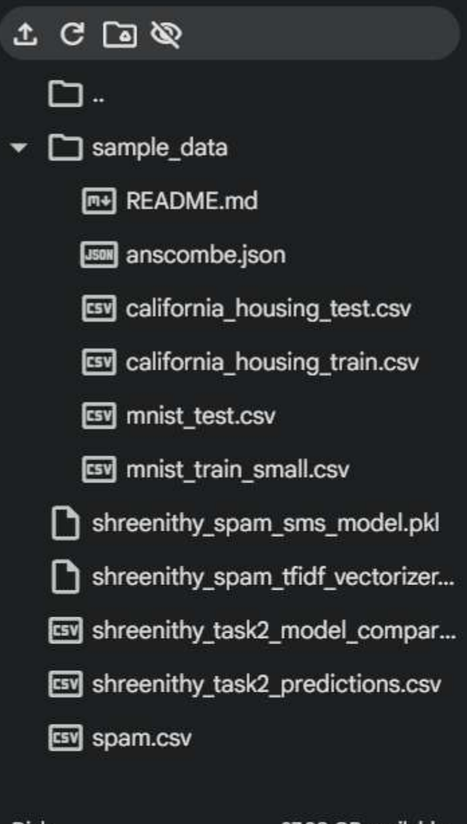
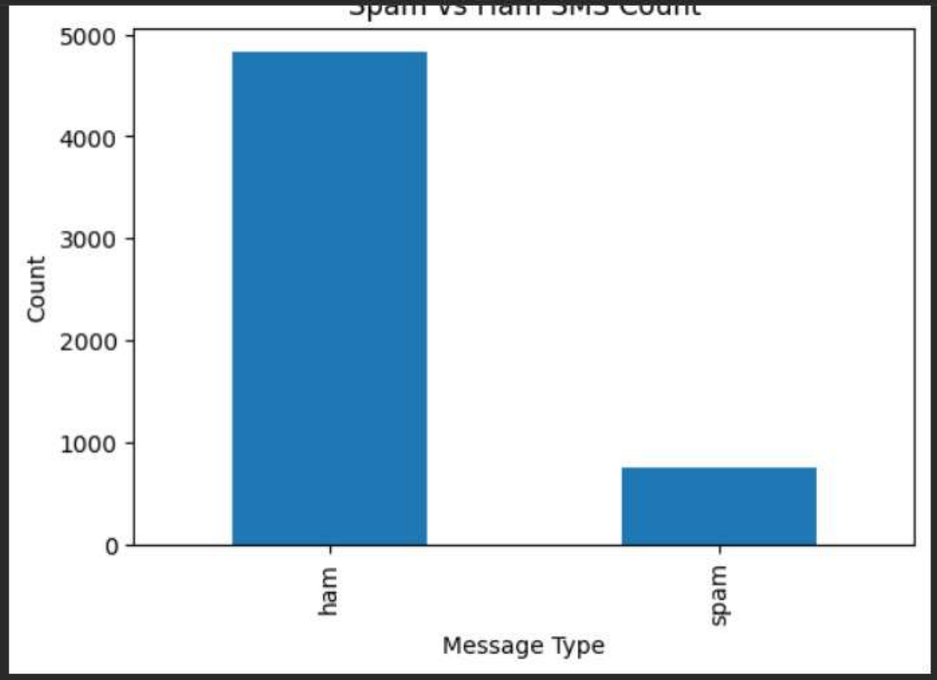



```
Final Dataset:
  Label                                            Message
0   ham  Go until jurong point, crazy.. Available only ...
1   ham                      Ok lar... Joking wif u oni...
2  spam  Free entry in 2 a wkly comp to win FA Cup fina...
3   ham  U dun say so early hor... U c already then say...
4   ham  Nah I don't think he goes to usf, he lives aro...

Dataset Shape: (5572, 2)

Missing Values:
Label      0
Message    0
dtype: int64

Spam and Ham Count:
Label
ham     4825
spam     747
Name: count, dtype: int64

Average Message Length:
Label
ham      71.023627
spam    138.866131
Name: Message_Length, dtype: float64
```



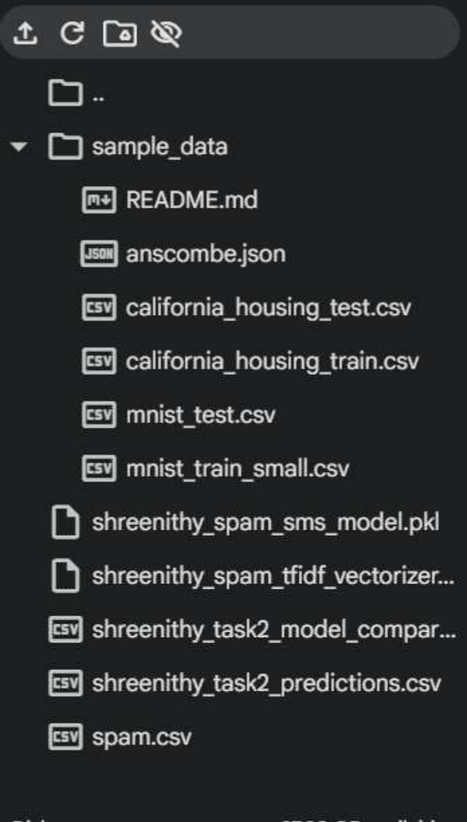
Files





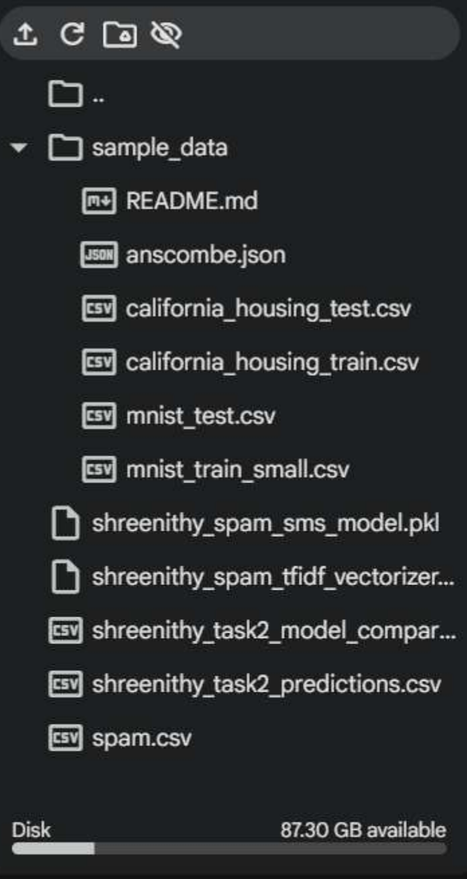

```
Original Message Example:
Go until jurong point, crazy.. Available only in bugis n great world la e buffet... Cine there got amore wat...

Cleaned Message Example:
go until jurong point crazy available only in bugis n great world la e buffet cine there got amore wat

Training Size: 4457
Testing Size: 1115
```

Files




```
Training Size: 4457
Testing Size: 1115

TF-IDF Vectorization completed!
Vector Shape: (4457, 8000)

====================================
Training Model: Naive Bayes
====================================
Accuracy: 0.9641255605381166
Precision: 0.990990990990991
Recall: 0.738255033557047
F1 Score: 0.8461538461538461

Classification Report:
              precision    recall  f1-score   support

         Ham       0.96      1.00      0.98       966
        Spam       0.99      0.74      0.85       149

    accuracy                           0.96      1115
   macro avg       0.98      0.87      0.91      1115
weighted avg       0.97      0.96      0.96      1115


====================================
Training Model: Logistic Regression
====================================
Accuracy: 0.9668161434977578
Precision: 0.9912280701754386
Recall: 0.7583892617449665
F1 Score: 0.8593155893536122
```

Files




```
Classification Report:
              precision    recall  f1-score   support

         Ham       0.96      1.00      0.98       966
        Spam       0.99      0.76      0.86       149

    accuracy                           0.97      1115
   macro avg       0.98      0.88      0.92      1115
weighted avg       0.97      0.97      0.96      1115


====================================
Training Model: Linear SVM
====================================
Accuracy: 0.9847533632286996
Precision: 0.9852941176470589
Recall: 0.8993288590604027
F1 Score: 0.9403508771929825

Classification Report:
              precision    recall  f1-score   support

         Ham       0.98      1.00      0.99       966
        Spam       0.99      0.90      0.94       149

    accuracy                           0.98      1115
   macro avg       0.98      0.95      0.97      1115
weighted avg       0.98      0.98      0.98      1115


====================================
```

Files
..
sample_data
README.md
anscombe.json
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california_housing_train.csv
mnist_test.csv
mnist_train_small.csv
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shreenithy_spam_tfidf_vectorizer...
shreenithy_task2_model_compar...
shreenithy_task2_predictions.csv
spam.csv
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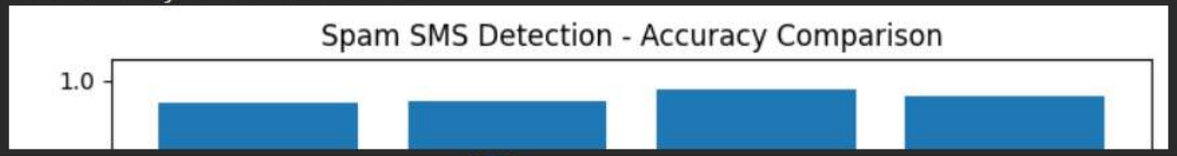
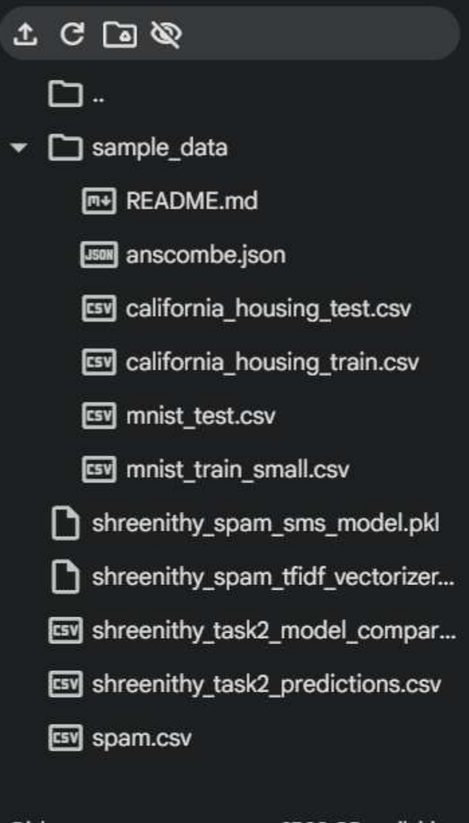
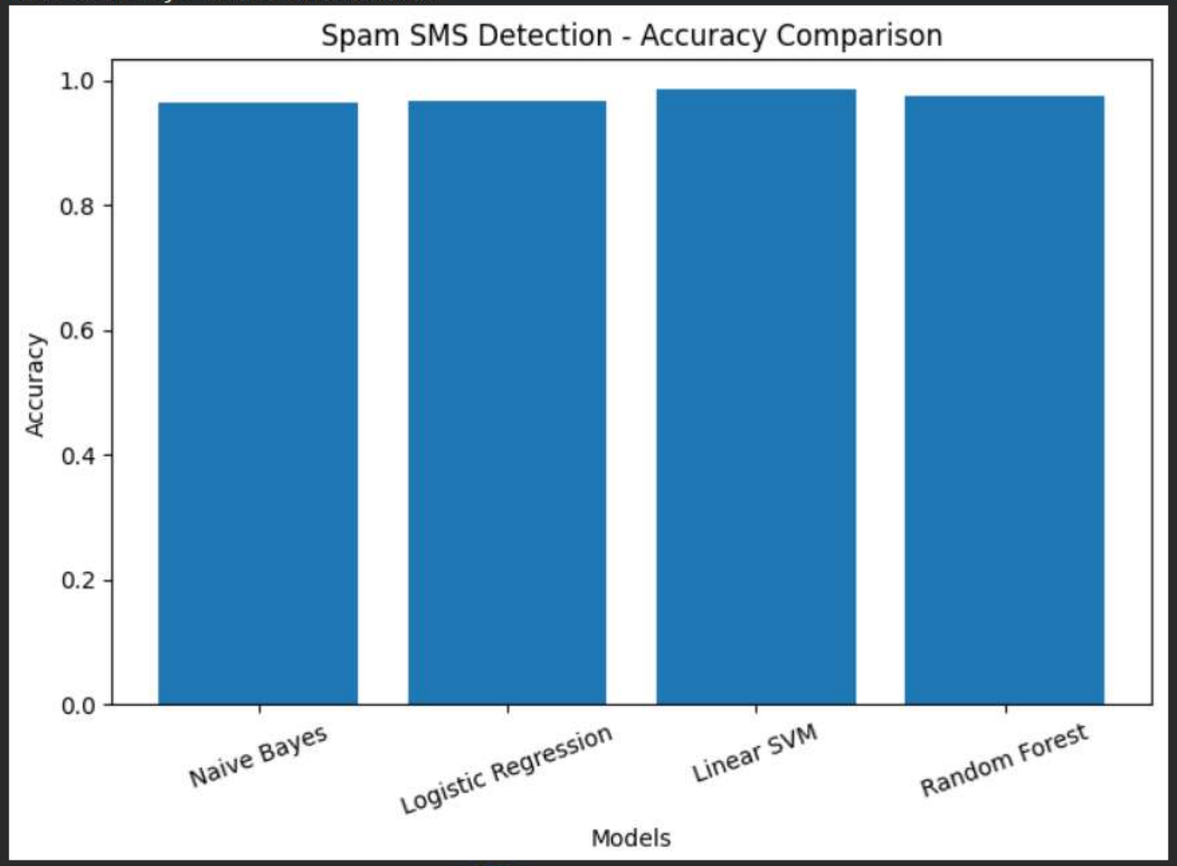
```
Training Model: Random Forest
==================================
Accuracy: 0.9757847533632287
Precision: 1.0
Recall: 0.8187919463087249
F1 Score: 0.9003690036900369

Classification Report:
              precision    recall  f1-score   support

         Ham       0.97      1.00      0.99       966
        Spam       1.00      0.82      0.90       149

    accuracy                           0.98      1115
   macro avg       0.99      0.91      0.94      1115
weighted avg       0.98      0.98      0.97      1115


========== MODEL COMPARISON ==========
                     Accuracy  Precision    Recall  F1 Score
Naive Bayes          0.964126   0.990991  0.738255  0.846154
Logistic Regression  0.966816   0.991228  0.758389  0.859316
Linear SVM           0.984753   0.985294  0.899329  0.940351
Random Forest        0.975785   1.000000  0.818792  0.900369

Best Model: Linear SVM
Best Accuracy: 0.9847533632286996
```



Files




```
Best Model: Linear SVM
Best Accuracy: 0.9847533632286996
```



Movie_Genre_Classification - Co | CodSoft_Task3_CustomerChurn | Spam_SMS_Detection.ipynb - C | Ask Gemini

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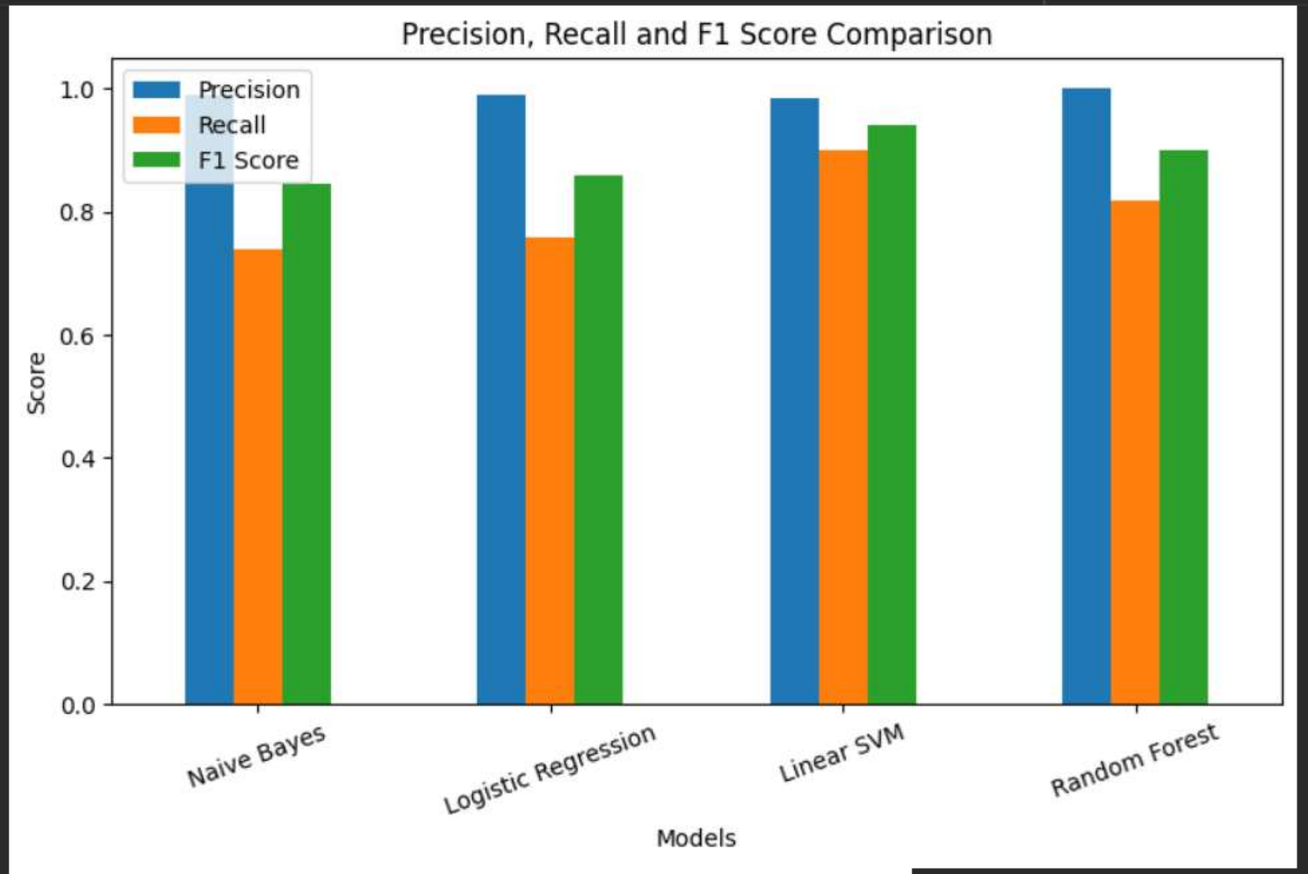

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Python 3

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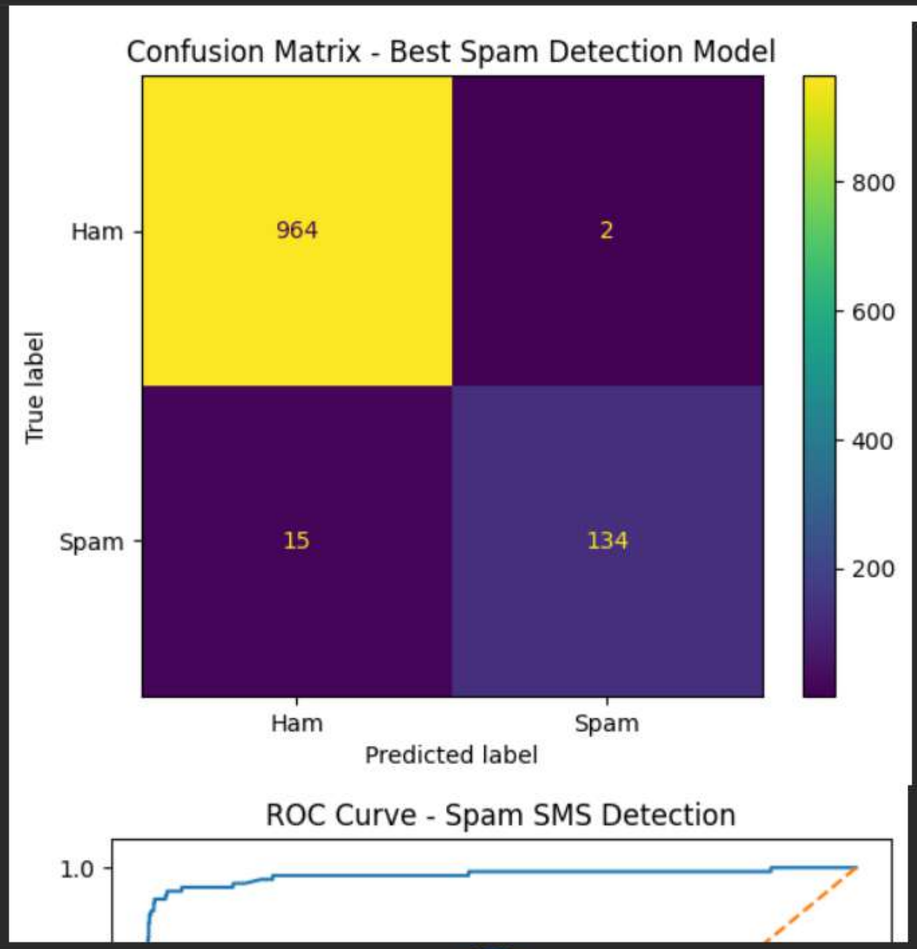

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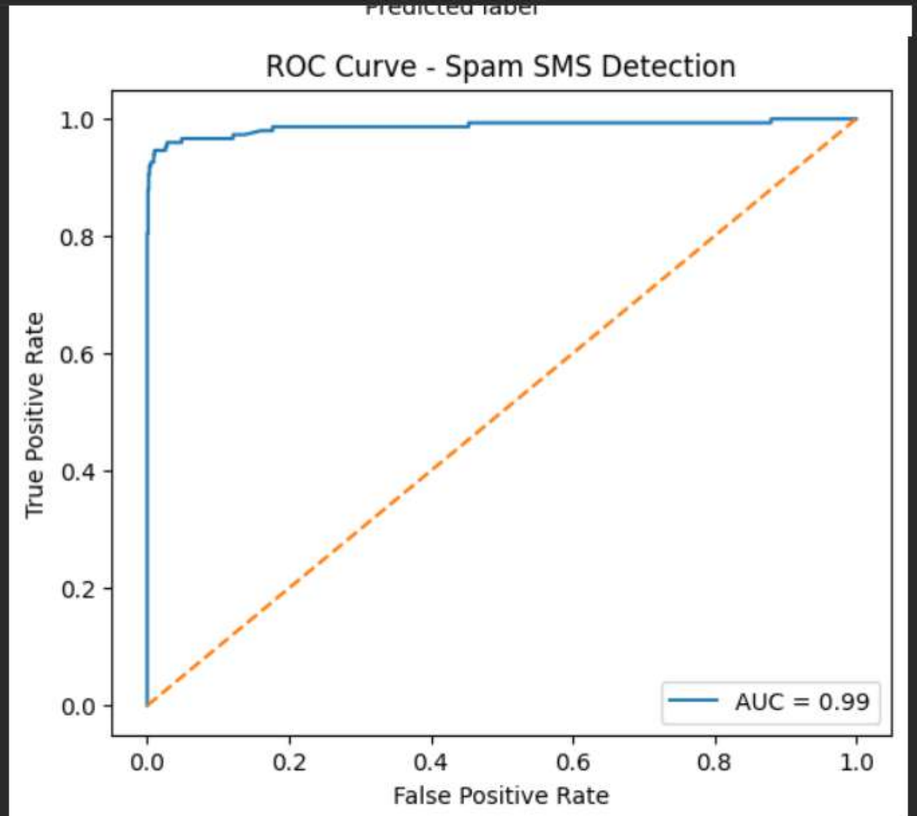

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```
Prediction CSV saved successfully!

========== SAMPLE SMS PREDICTIONS ==========

SMS: Congratulations! You have won a free iPhone. Click the link now.
Prediction: Spam Message

SMS: Hey, are you coming to college tomorrow?
Prediction: Ham Message

SMS: You won 50000 rupees cash prize. Claim immediately.
Prediction: Spam Message

SMS: Your OTP is 4589. Do not share it with anyone.
Prediction: Ham Message

SMS: Free recharge offer available only today. Click here.
Prediction: Ham Message

SMS: Call me when you reach home.
Prediction: Ham Message

SMS: URGENT! Your account has been blocked. Verify now.
Prediction: Spam Message

SMS: Good morning, submit your assignment today.
Prediction: Ham Message

Enter any SMS message to check Spam or Ham: Call me when you reach home.

Your SMS:
Call me when you reach home.
```

Variables Terminal

Python 3

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- shreenithy_task2_predictions.csv
- spam.csv

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```
Your SMS:
Call me when you reach home.

Prediction Result:
Ham Message

Model and vectorizer saved successfully!

========== TASK 2 COMPLETED ==========
Project Name: Spam SMS Detection
Best Model: Linear SVM
Best Accuracy: 0.9847533632286996

Models Used:
- Naive Bayes
- Logistic Regression
- Linear SVM
- Random Forest

Task 2 completed successfully!
```

Double-click (or enter) to edit